

STATISTICS 652

LECTURE 3-26

CONVEXITY AND CONCAVITY

DEF 3.1. A SET $(H) \subset \mathbb{R}^k$ IS CONVEX IFF FOR ALL $\theta_0 \in (H), \theta_1 \in (H)$, ITS
CONVEX COMBINATION $(1-\lambda)\theta_0 + \lambda\theta_1 \in (H)$ FOR ALL $0 \leq \lambda \leq 1$

NOTE: $(H) \subset \mathbb{R}$ IS CONVEX IFF IT IS AN INTERVAL (OPEN, CLOSED, HALF-OPEN, UNBO)

NOTE: $\partial(H)$: BOUNDARY SET, $(H)_I$ INTERIOR, $(H)_C$ CLOSURE

DEF 3.2. FOR $(H) \subset \mathbb{R}^k$ CONVEX, A FUNCTION $g: (H) \rightarrow \mathbb{R}$ IS CONCAVE IFF $\forall \theta_0, \theta_1 \in (H)$

$$g((1-\lambda)\theta_0 + \lambda\theta_1) \geq (1-\lambda)g(\theta_0) + \lambda g(\theta_1) \quad *$$

FOR ALL $0 \leq \lambda \leq 1$. IT IS STRICTLY CONCAVE IF STRICT INEQUALITY ALWAYS HOLDS,
WHEN $\theta_0 \neq \theta_1$.

NOTE: FOR CONVEX AND STRICTLY CONVEX FUNCTION, REPLACE $\geq$ WITH $\leq$ IN $*$.

LEMMA 3.3. SUPPOSE $g: (H) \rightarrow \mathbb{R}$ IS CONCAVE, THEN

- ANY LOCAL MAXIMUM IS A GLOBAL MAXIMUM
- THERE ARE NO LOCAL MINIMA OR INFLECTION POINTS ON $(H)_I$.
- SET OF ALL GLOBAL MAXIMUM IS EITHER CONVEX OR EMPTY
- IF (H) IS BOUNDED, THEN A MAXIMUM EXISTS ON $(H)_C$. HENCE, IF (H) IS CLOSED AND BOUNDED, A MAXIMUM EXISTS ON (H) .
 - IF $(H) = \mathbb{R}^k$, THEN A MAXIMUM EXISTS IFF g IS COERCIVE, I.E. $g(\theta) \rightarrow -\infty$ AS $\|\theta\| \rightarrow \infty$.

OTHER BASIC PROPERTIES

- g CONCAVE (STRICT) $\Leftrightarrow h = -g$ CONVEX (STRICT)

MAKE MODIFICATIONS IN LEMMA 3.3:

$$\text{MAX} \leftrightarrow \text{MIN}$$

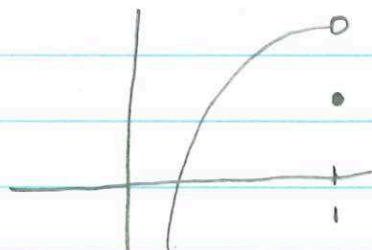
$$\text{MIN} \leftrightarrow \text{MAX}$$

$$-\infty \leftrightarrow \infty$$

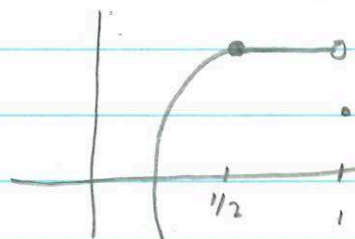
- g IS LINEAR IFF g IS CONCAVE AND CONVEX
- SUM CONCAVE (CONVEX) FUNCTIONS IS CONCAVE (CONVEX) AND STRICTLY SO IF ONE OF THE SUMMANDS IS STRICTLY CONCAVE (CONVEX),

- CONCAVE OR CONVEX FUNCTION ON CONVEX $(H) \subset \mathbb{R}^k \Rightarrow$ CONTINUOUS ON INTERIOR

EXAMPLE:



NO MAXIMUM ON $(H)_c$



SET OF MAXIMA IS CONVEX BUT NOT CLOSED

$$\text{MAX} = \{ \theta \mid \frac{1}{2} \leq \theta < 1 \}$$

$$(H) = \{ 0 < \theta \leq 1 \}$$

$$(H)_c = \{ 0 \leq \theta \leq 1 \}$$

$$(H)_I = \{ 0 < \theta < 1 \}$$

RECALL:

EXPONENTIAL FAMILY, NATURAL PARAMETERIZATION

$$f(\underline{x}; \underline{\eta}) = \exp \left\{ \sum_{j=1}^k \eta_j T_j(\underline{x}) + d_0(\underline{x}) + S(\underline{x}) \right\} I_A(\underline{x}) \quad \underline{\eta} \in \bar{\eta}$$

$$= \exp \{ \underline{\eta}' \underline{T}(\underline{x}) + d_0(\underline{x}) + S(\underline{x}) \} I_A \quad \underline{\eta} \in \mathbb{R}^k \quad \underline{T}: \mathbb{R}^m \rightarrow \mathbb{R}^k$$

$$\bar{\eta} = \left\{ \underline{\eta} = (\eta_1, \dots, \eta_k) \mid \underline{\eta} = (c_1 \theta_1, \dots, c_k \theta_k), \theta \in \Theta \right\}$$

RECALL: η is a 1-1 function of θ (by identifiability)

Lemma 2.4: $\underline{\eta}_0, \underline{\eta}_1 \in \bar{\eta} \Rightarrow f(\underline{x}; \underline{\eta})$ is a valid pdf (pmf) for $\underline{\eta} = (1-\lambda)\underline{\eta}_0 + \lambda\underline{\eta}_1$,
i.e. convex closure of $\bar{\eta}$ is a valid parameter space.

SEE EXAMPLE: LECTURE 2, PAGE 6

LEMMA 3.4: UNIQUENESS OF NATURAL PARAMETERIZATION UNDER AFFINE TRANS

$\bar{\eta}_0$ represents a natural parameterization for an exponential family iff $\bar{\eta}_1 = \{ \underline{\eta}_1 \mid \underline{\eta}_1 = A \underline{\eta}_0 + \underline{b}, \underline{\eta}_0 \in \bar{\eta}_0 \}$
represents a natural parameterization for any $\underline{b} \in \mathbb{R}^k$, A non-singular, $k \times k$

PROOF: HW (NOTE: DEFINITION OF $T_j(\underline{x})$ and $d_0(\underline{x})$ CHANGES.)

PROOF OF LEMMA 2.4*. CONTINUOUS CASE, SHOW

$$\int_A \exp \left\{ \sum_{j=1}^k \eta_j T_j(\underline{x}) + S(\underline{x}) \right\} d\underline{x} < \infty \quad \text{for } \underline{\eta} = (1-\lambda)\underline{\eta}_0 + \lambda\underline{\eta}_1$$

when it holds for $\underline{\eta}_0$ and $\underline{\eta}_1$.

$$\begin{aligned} q(\underline{\eta}) &= \int_A \exp \left\{ (1-\lambda) \sum_{j=1}^k \eta_{0,j} T_j(\underline{x}) + \lambda \sum_{j=1}^k \eta_{1,j} T_j(\underline{x}) + S(\underline{x}) \right\} d\underline{x} \\ &= \int_A (1-\lambda) \exp \left\{ \sum_{j=1}^k \eta_{0,j} T_j(\underline{x}) + S(\underline{x}) \right\} + \lambda \exp \left\{ \sum_{j=1}^k \eta_{1,j} T_j(\underline{x}) + S(\underline{x}) \right\} d\underline{x} \quad \begin{array}{l} \text{since} \\ C^4 \text{ is convex} \end{array} \\ &= (1-\lambda) q(\underline{\eta}_0) + \lambda q(\underline{\eta}_1) < \infty. \end{aligned}$$

COROLLARY 3.5. $g(\eta) = e^{-d_0(\eta)}$ is STRICTLY CONVEX.

PROOF: HW

THEOREM 3.6. Let $X \sim f_\eta$, $\eta \in \bar{\eta}$ be an exponential family in natural form with $\bar{\eta}$ being convex. THEN $L(\eta; \bar{X})$

IS STRICTLY LOG-CONCAVE IN $\eta \in \bar{\eta}$, i.e.

$\ell(\eta; \bar{X}) = \log L(\eta; \bar{X})$ IS STRICTLY CONCAVE.

MOTIVATION:

$X_1, \dots, X_n$ iid $\text{Gamma}(\alpha, \beta)$ $\alpha > 0, \beta > 0$

$$\ell(\alpha, \beta; \mathbf{x}) = (\alpha-1) \sum_{i=1}^n \log x_i - \frac{1}{\beta} \sum_{i=1}^n x_i - n \log \Gamma(\alpha) - n \alpha \log \beta.$$

CASE I: α -known

$$\partial \ell / \partial \beta = \frac{\sum x_i}{\beta^2} - \frac{n\alpha}{\beta} \Rightarrow \hat{\beta} = \frac{\bar{x}}{\alpha} \quad \text{ONLY CRITICAL POINT} \Rightarrow \text{UNIQUE MLE}$$

WHY?

CASE II: β -known

$$\partial \ell / \partial \alpha = \sum_{i=1}^n \log x_i - n \Psi(\alpha) - n \log \beta \quad \text{where } \Psi(\alpha) = \Gamma'(\alpha) / \Gamma(\alpha) \quad \text{Digamma FUNCTION}$$

$$= 0 \Rightarrow \Psi(\alpha) = \overline{\log x} - \log \beta \quad \text{ESTIMATING EQUATION}$$

BY

EXP. FAMILY: UNIQUE SOLUTION OR NO SOLUTION, i.e., $\hat{\alpha} = 0$ or $\hat{\alpha} = \infty$

ALTERNATIVE: SHOW $\Psi'(\alpha) = \text{VAR}(\log X)$ for $X \sim \text{Gamma}(\alpha, 1)$. HW

$$\text{HENCE } \partial^2 \ell / \partial \alpha^2 = -n \Psi'(\alpha) < 0$$

CASE III: α, β UNKNOWN

$$\text{ESTIMATING EQUATIONS: } \hat{\alpha} \hat{\beta} = \bar{x} \quad \text{and} \quad \Psi(\hat{\alpha}) = \overline{\log x} - \log \hat{\beta}$$

$$\Leftrightarrow \hat{\beta} = \bar{x} / \hat{\alpha} \quad \text{and} \quad \Psi(\hat{\alpha}) - \log(\hat{\alpha}) = \overline{\log x} - \log \bar{x}$$

NOTE: UNIQUE OR NO SOLUTION

NUMERICALLY FIND A SOLUTION $\Rightarrow$ UNIQUE SOLUTION

HW: For $X_1, \dots, X_n$ i.i.d. $\text{Gamma}(d, 1)$, show MLE for d exists and is unique.
HINT: STERLING'S APPROXIMATION.

PROOF OF THEOREM 3.6, CONTINUOUS CASE (DISCRETE IS ANALOGOUS)

$$\begin{aligned} \ell(\eta; \underline{n}) &= \sum_{j=1}^k \eta_j T_j(\underline{n}) + d_0(\eta) + S(\underline{n}) \\ &= \eta' \underline{T}(\underline{n}) + d_0(\eta) + S(\underline{n}) \end{aligned}$$

Now: $\eta' \underline{T}(\underline{n}) + S(\underline{n})$ is linear in η , hence concave

THUS: IT IS SUFFICIENT TO SHOW $d_0(\eta)$ IS STRICTLY CONCAVE (WHY?)

$$\text{i.e. } d_0((1-\lambda)\eta_0 + \lambda\eta_1) > (1-\lambda)d_0(\eta_0) + \lambda d_0(\eta_1) \quad 0 < \lambda < 1$$

$$\text{RECALL: } d_0(\eta) = -\log \int_A \exp\{\eta' \underline{T}(\underline{n}) + S(\underline{n})\} d\underline{n}$$

$$\text{so } d_0((1-\lambda)\eta_0 + \lambda\eta_1)$$

$$= -\log \left\{ \int_A \exp\{(1-\lambda)(\eta_0 - \eta_1)' \underline{T}(\underline{n})\} \exp\{\eta_1' \underline{T}(\underline{n}) + d_0(\eta_1) + S(\underline{n})\} d\underline{n} e^{-d_0(\eta_1)} \right\}$$

$$= -\log E_{\eta_1} \left[\exp\{(1-\lambda)(\eta_0 - \eta_1)' \underline{T}(\underline{n})\} \right] + d_0(\eta_1)$$

$$> -\log \left\{ E_{\eta_1} [\exp\{(\eta_0 - \eta_1)' \underline{T}(\underline{n})\}] \right\}^{1-\lambda} + d_0(\eta_1)$$

SINCE BY JENSEN'S INEQUALITY,

For STRICTLY CONVEX g : $E[g(Y)] > g(E(Y))$ (CONCAVE: ANALOGOUS)

PROVIDED $P(Y=y) \neq 1$ for all y .

NOTE: $g(y) = y^\beta$ is STRICTLY CONCAVE ON $y > 0$, for $0 < \beta < 1$.

$$= -(1-\lambda) \log \left[\int_A \exp\{(\eta_0 - \eta_1)' \underline{T}(\underline{n})\} \exp\{\eta_1' \underline{T}(\underline{n}) + d_0(\eta_1) + S(\underline{n})\} d\underline{n} \right] + d_0(\eta_1)$$

$$= -(1-\lambda) \log \left[\int_A \exp\{\eta_0' \underline{T}(\underline{n}) + S(\underline{n})\} \cdot \exp\{\eta_1' \underline{T}(\underline{n}) + d_0(\eta_1)\} d\underline{n} \right] + d_0(\eta_1)$$

$$= (1-\lambda) d_0(\eta_0) + \lambda d_0(\eta_1) \quad \square$$